DevOps

Crontab and systemD

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What is Crontab?

- **Crontab** is a system process that will automatically perform tasks as per the specific schedule. It is a set of commands that are used for running regular scheduling tasks.

- **Why use Crontab?**
  - Helps OS to take a scheduled backup of log files or database.
  - Delete old log files
  - Archive and purge database tables
  - Send out any notification email such as Newsletters, Password expiration email
  - Regular clean-up of cached data
  - It is used to automate system maintenance

How to use crontab

- **Crontab guideline:** https://crontab.guru/

```
.---------------- minute (0 - 59)
|  .------------- hour (0 - 23)
|  |  .---------- day of month (1 - 31)
|  |  |  .------- month (1 - 12) OR jan,feb,mar,apr ...
|  |  |  |  .---- day of week (0 - 6) (Sunday=0 or 7)  OR sun,mon,tue,wed,thu,fri,sat
|  |  |  |  |
*  *  *  *  *  command/script to be executed
```

- `$ cat /etc/crontab # Show how to use crontab`
- `$ crontab -e # Create cron job`
- `$ crontab -l # Check and list cron job`
- `$ crontab -i -r # prompt before deleting user's crontab`

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Specifying multiple values in a field

- The asterisk (*) operator specifies all possible values for a field. e.g. every hour or every day.
- The comma (,) operator specifies a list of values, for example: "1,3,4,7,8".
- The dash (-) operator specifies a range of values, for example: "1-6", which is equivalent to "1,2,3,4,5,6".
- The slash (/) operator, can be used to skip a given number of values. For example, "*/3" in the hour time field is equivalent to "0,3,6,9,12,15,18,21"; "*" specifies 'every hour' but the "/3" means that only the first, fourth, seventh...and such values given by "*" are used.

Example

- `$ crontab -e`
- Note: Use space to separate each block

```
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   commandi
*/3 * * * * /home/seng/test-crontab.sh
```

- */3 : It will execute script in every 3 minutes.
- `$ sudo vim test-crontab.sh`

```
echo "Today $(date)" >> crontab-date.txt
```

- `$ sudo chmod +x test-crontab.sh`

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Log rotation

- **Log rotation** is the process of archiving, compressing, and periodically removing old log files generated by an application, system or service in order to conserve disk space and maintain a manageable set of logs. The objective of log rotation is to prevent a single log file from consuming all available disk space and keep logs organized, easily accessible and readily available for analysis.

Log rotation

- `$ sudo vim /etc/logrotate.d/vault`

```
/var/log/vault_audit.log {
   rotate 7
   size 10M
   daily
   #Do not execute rotate if the log file is empty.
   notifempty
   missingok
   compress
   #Set compress on next rotate cycle to prevent entry loss when performing compression.
   delaycompress
   copytruncate
   extension log
   dateext
   dateformat -%Y-%m-%d-%H%M%S.
   su root adm
   #olddir /mnt/test
   #sharedscripts
   #postrotate
   #       /scripts/backuplog.sh
   #endscript
}
```

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Log rotation

- Run Manual Test

```
$ logrotate -f /etc/logrotate.d/vault
=== OR ===
$ logrotate -f /etc/logrotate.conf
```

SystemD Service

- SystemD Service put a service is a "background process" that is started or stopped based on certain circumstances. You are not required to manually start and/or stop it.
- `$ cd /etc/systemd/system`
- `$ sudo vim my-reboot.service`

```
[Unit]
Description=REBOOT-WITH-SCRIPT
After=network.target
[Service]
ExecStart=/opt/reboot/reboot.sh
[Install]
WantedBy=default.target
```

SystemD Service

- $ cd /opt/reboot
- $ sudo vim reboot.sh

```
#!/bin/bash
echo "$(date "+%F@%T")" >> /opt/reboot/reboot.txt
```

- $ sudo chmod +x reboot.sh
- $ sudo systemctl daemon-reload
- $ sudo systemctl start my-reboot.service
- $ sudo systemctl enable my-reboot.service
- $ sudo systemctl status my-reboot.service # inactive is okay
- $ sudo reboot

Target Units

- There are many target units available in systemd, and the exact set of target units can vary depending on the specific distribution and configuration of the system. However, some common target units in systemd include:
- multi-user.target: A multi-user system with network support
- graphical.target: A system with a graphical user interface
- default.target: The default target for the system, reached when the system has finished booting
- emergency.target: A single-user system used for emergency and rescue purposes

Target Units

- rescue.target: A rescue shell that provides a single-user system with limited resources
- single.target: A single-user system without network support
- shutdown.target: A target reached when the system is shutting down
- poweroff.target: A target reached when the system is powered off
- reboot.target: A target reached when the system is rebooting

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Unit vs Install

- A unit refers to a configuration file that describes a specific aspect of the system, such as a service, a socket, a device, or a mount point. Each unit file contains information about how the component should be started, stopped, and restarted, as well as any dependencies and dependencies of other units.
- On the other hand, "install" refers to the process of making a unit file available to systemd so that it can be managed by the system. When a unit file is installed, it is typically placed in one of the unit file directories, such as /usr/lib/systemd/system or /etc/systemd/system, and it is linked to a target unit, which specifies when the unit should be started.

Unit vs Install

- The [Install] section of a unit file contains information about how the unit should be installed and enabled, including the target unit that the unit should be linked to. The "WantedBy" directive in the [Install] section specifies the target unit that the service should be started with.

Thank you

Begin with the theory, Start with the practical